

Statistics

Df :-

→ Science of collecting, organizing and analyzing data.

⇒ for decision making

Types of Statistics :-

1- Descriptive Statistics

Involves methods for summarizing and organizing data to make it understandable. This type of statistics helps to describe the basic features of the data in a study.

1. Measure of central tendency
[mean, mode, median]

2. Measure of Dispersion
[variance, sd]

3. Data distribution

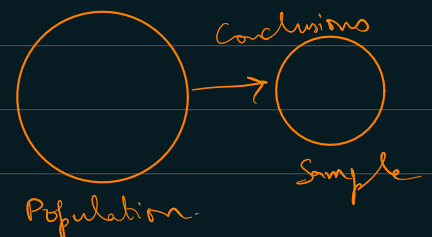
[Histograms, bar plot, pie chart, PMF, PDF]

4. Summary statistics

[Five number summary, Q_1 , Q_2 , Q_3 , maximum]

2- Inferential Statistics

Involves methods for making predictions or inferences about a population based on a sample of data. It allows for hypothesis testing, estimation and drawing conclusions.



1. Hypothesis testing

2. P value

3. Confidence interval

4. Statistical Analysis test

[Z test, t test, ANOVA, Chi Square]

Population

→ Entire set

Characteristics :-

1. Complete set: Contains all the observations of interest

2. Parameter: A numerical value

Summarizing the entire pop'n

a) Population mean (μ)

b) Population variance (σ^2)

Sample data

→ Subset of population

1. Subset: Represents portion of pop'n

2. Statistic: A numerical value

Summarizing the sample

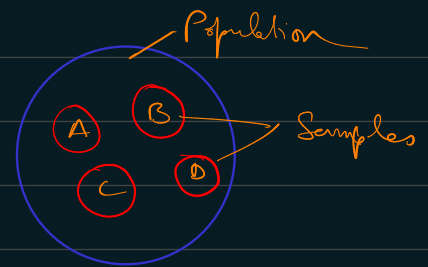
a) Sample mean

b) Sample variance

3. Random sampling: Should be random

* Types of Sampling techniques:-

- 1- Probability Sampling
- 2- Non-probability Sampling



① Probability Sampling:-

a) Simple Random Sampling:-

Every member of the population has an equal chance of being selected.

Ex:- Selecting people randomly.
Draw names randomly

b) Systematic Sampling:-

Select every n^{th} member of the pop. after a random starting point.

Ex:- Feedback survey from customers

c) Stratified Sampling:-

Divide the pop. into strata (groups) based on specific characteristics & then randomly sample from each strata

Ex:- Divide employees in a company by dept. & then select a proportional number from each dept to form a survey sample.

Ex:- Age: < 12 $12 - 18$ > 18

d) Cluster Sampling:-

Divide the pop. into clusters, randomly selecting clusters, then sampling all the members from the selected clusters.

Ex:- Randomly selecting several schools in a dist. & then surveying all teachers within those schools.

e) Multi Stage Sampling:-

Combining several sampling methods - Usually involves selecting

* Types of Data :-



* Scales of measurement of Data

1. Nominal Scale
2. Ordinal Scale
3. Interval
4. Ratio

① Nominal Scale :-

This scale classifies data into distinct categories that don't have an intrinsic order.

→ Qualitative / Categorical data

Characteristics:-

- i. Data is categorized based on labels, names or qualities
- ii. Categories are mutually exclusive. Gender → M // F
- iii. No logical order (no ranking)

Ex- Colours, Genders, Cuisines

② Ordinal Scale:-

This scale classifies the data into categories that can be ranked/ordered.

Characteristics:-

- i. Data is categorized and ranked in a specific order.
- ii. The interval b/w ranks are necessarily equal.

Ex: Education level	Rank	Customer feedback
HS	1	Very satisfied
Bachelor	2	Satisfied
Masters	3	neutral
Doctorate	4	not satisfied

③ Interval Scale:-

The interval scale not only categorizes & orders but also specifies the exact difference b/w intervals. It lacks the zero point.

Characteristics:-

- i. Data is ordered with consistent interval b/w values.
- ii. It allows for meaningful comparison of differences (Ratio cannot be measured).
- iii. No true zero point.

Ex: Temperature in F

10°F, 20°F, 30°F

20 - 10

30 - 10

0°F $\neq$ No temperature

Ex: Calendar years

④ Ratio Scale:-

- i. The order matters.
- ii. Differences are measurable (Ratio can be measured).
- iii. Contains 0 static point.

